INTERN PERFORMANCE ANALYTICS AND INSIGHTS DASHBOARD

SPRINT 2 – EXPLORATORY DATA ANALYSIS (EDA)

(High Level Documentation)

1. Statistical Analysis:

The dataset was analyzed using statistical measures such as average, total, and distribution to understand overall intern performance. Key metrics like average performance rate, attendance, productivity, and total tasks completed were calculated.

2. Trend and Pattern Identification:

Relationships between different variables were explored to identify meaningful patterns. The analysis focused on understanding how factors such as attendance, training hours, productivity, and feedback influence intern performance.

3. Data Visualization:

Various visualizations were created using Power BI to represent the data effectively. Charts such as bar charts, scatter plots, and pie charts were used to highlight patterns and comparisons across departments, categories, and performance levels.

4. Analysis Output and Preparation:

The analysis results were organized and prepared for further use in dashboard development. Insights derived from this stage serve as the foundation for KPI definition and visualization in the next sprint.

INTERN PERFORMANCE ANALYTICS AND INSIGHTS DASHBOARD

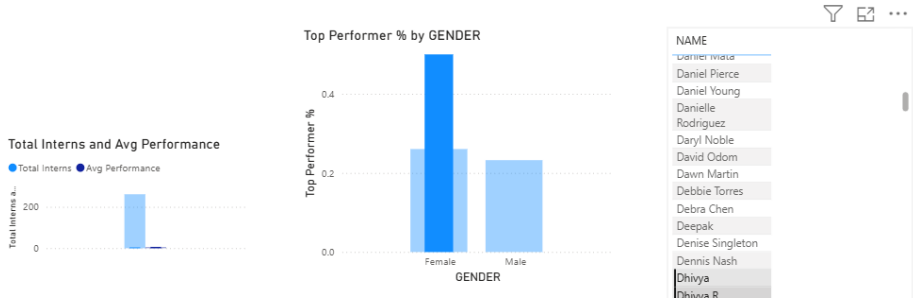
SPRINT 2 – EXPLORATORY DATA ANALYSIS (EDA)

(Low Level Documentation)

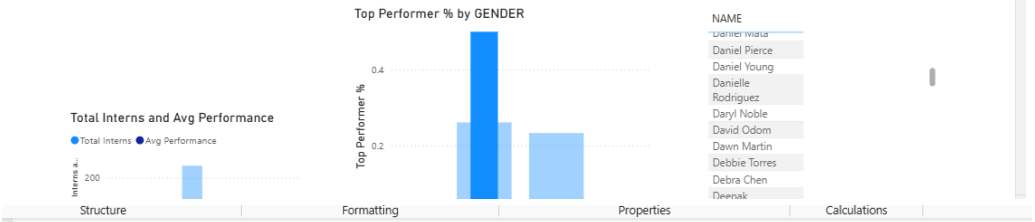
1. Statistical Analysis:

DAX measures were created in Power BI to calculate key metrics such as average performance, average attendance, average productivity, total interns, and total tasks completed. These measures helped in summarizing the dataset and understanding overall trends.

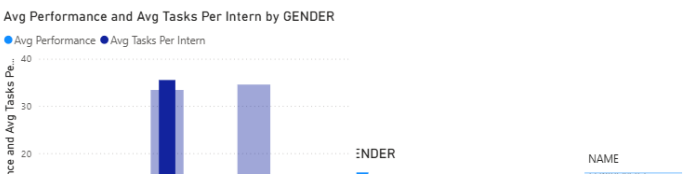
```
Top Performer % =
DIVIDE(
    CALCULATE(COUNT(Form_Responses[NAME]), Form_Responses[TOP_PERFORMER] = "Yes"),
    COUNT(Form_Responses[NAME])
)
```



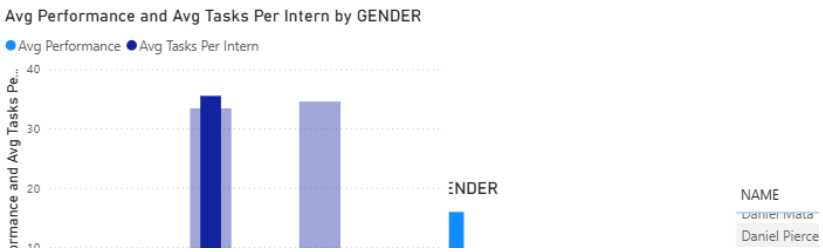

```
Total Interns = COUNT(Form_Responses[NAME])
```



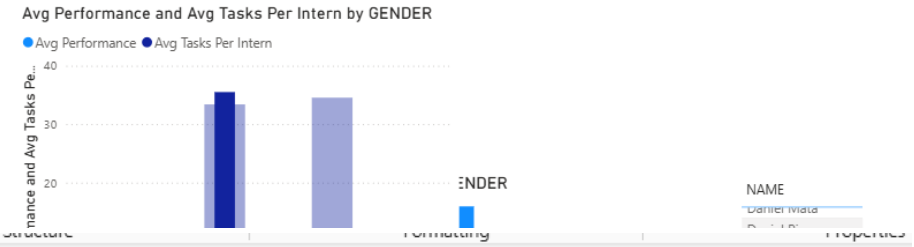

```
Avg Performance = AVERAGE(Form_Responses[PERFORMANCE_RATE])
```



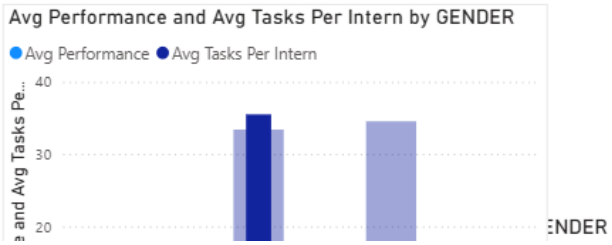
```
Avg Tasks Per Intern = AVERAGE(Form_Responses[TASK_COMPLETED])
```




```
Avg Productivity = AVERAGE(Form_Responses[PRODUCTIVITY_RATIO])
```

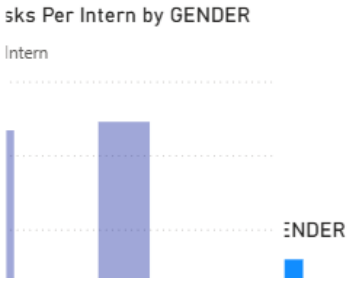


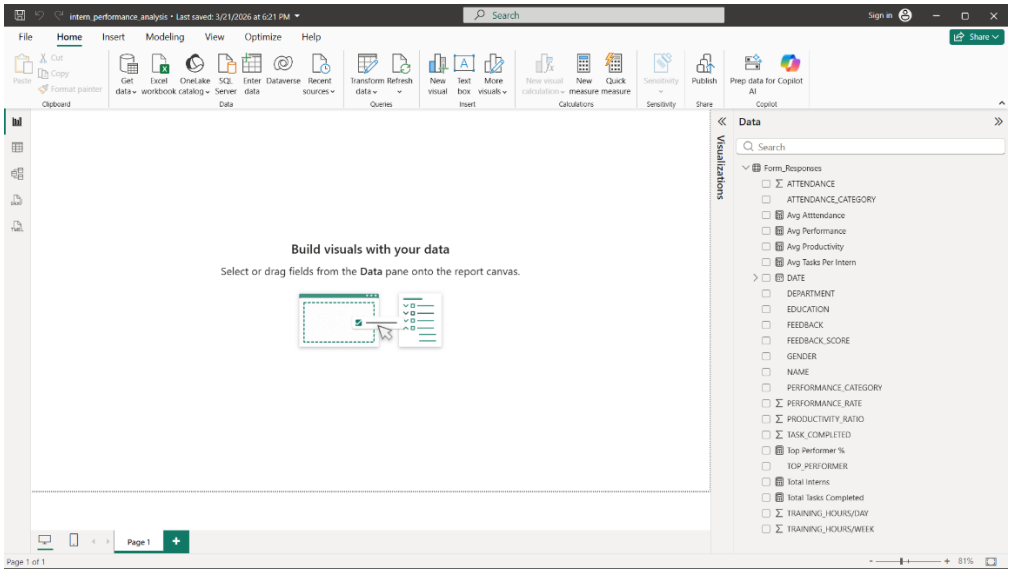

```
Avg Atttendance = AVERAGE(Form_Responses[ATTENDANCE])
```




```
Total Tasks Completed = SUM(Form_Responses[TASK_COMPLETED])
```

FEEDBACK_SCORE	Total Tasks Completed
Negative	9
Neutral	12
Positive	121
Total	**142**

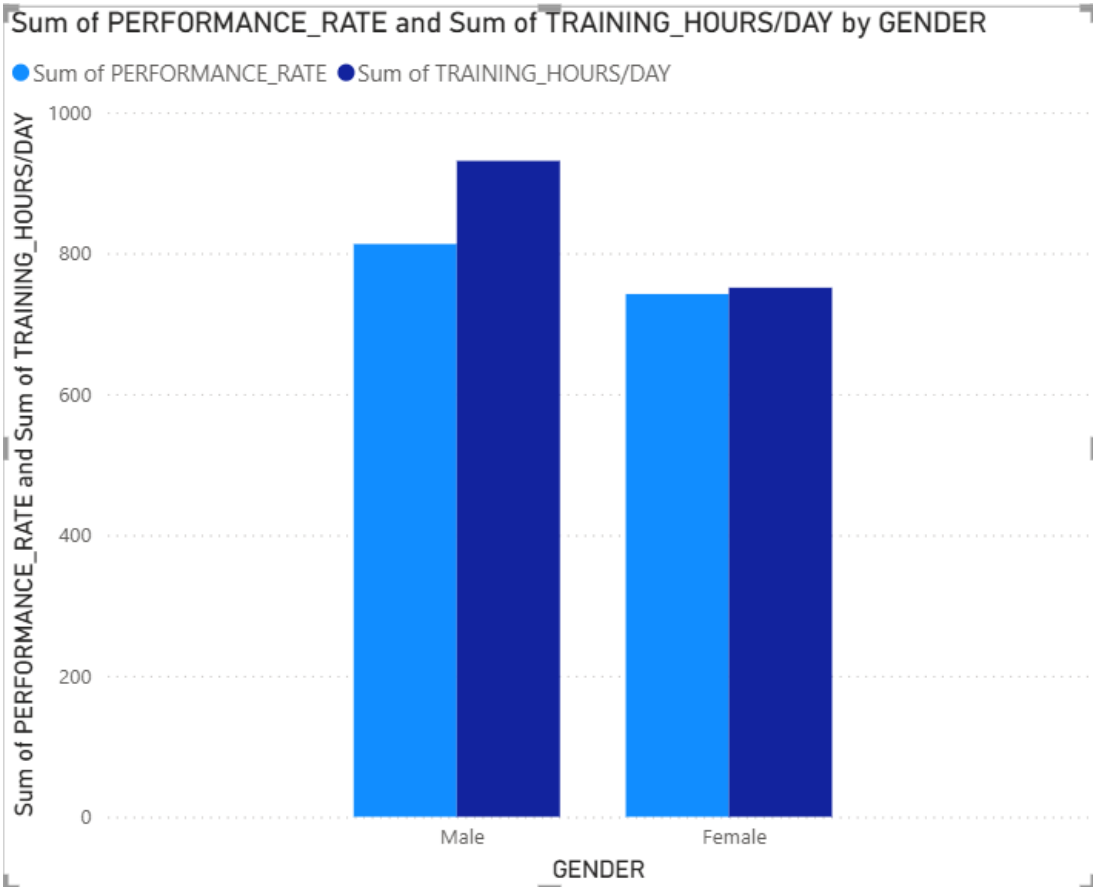




2. Trend and Pattern Identification:

Several relationships were analyzed to identify patterns in the dataset:

- Attendance vs Performance to check consistency impact
- Training Hours vs Performance to evaluate learning effect
- Productivity Ratio vs Performance to analyze efficiency
- Department-wise comparison to identify high-performing departments
- Feedback vs Performance to understand qualitative impact

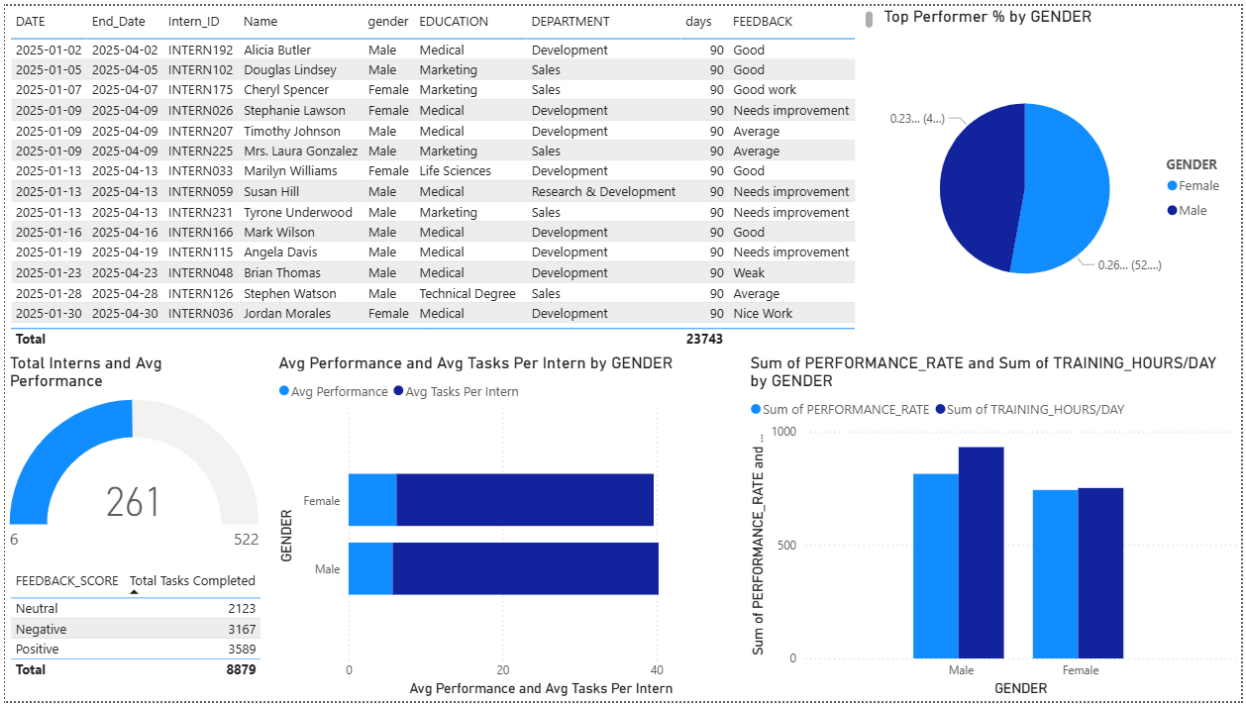


These analyses revealed strong correlations between performance and key influencing factors.

3. Data Visualization:

Visualizations were created in Power BI to support the analysis:

- Scatter Plot for Attendance vs Performance
- Bar Chart for Department vs Performance
- Column Chart for Training Hours vs Performance
- Scatter Plot for Productivity vs Performance
- Pie Chart for Performance Category Distribution



DATE	End_Date	Intern_ID	Name	gender	EDUCATION	DEPARTMENT	days	FEEDBACK
2025-01-02	2025-04-02	INTERN192	Alicia Butler	Male	Medical	Development	90	Good
2025-01-05	2025-04-05	INTERN102	Douglas Lindsey	Male	Marketing	Sales	90	Good
2025-01-07	2025-04-07	INTERN175	Cheryl Spencer	Female	Marketing	Sales	90	Good work
2025-01-09	2025-04-09	INTERN026	Stephanie Lawson	Female	Medical	Development	90	Needs improvement
2025-01-09	2025-04-09	INTERN207	Timothy Johnson	Male	Medical	Development	90	Average
2025-01-09	2025-04-09	INTERN225	Mrs. Laura Gonzalez	Male	Marketing	Sales	90	Average
2025-01-13	2025-04-13	INTERN033	Marilyn Williams	Female	Life Sciences	Development	90	Good
2025-01-13	2025-04-13	INTERN059	Susan Hill	Male	Medical	Research & Development	90	Needs improvement
2025-01-13	2025-04-13	INTERN231	Tyrone Underwood	Male	Marketing	Sales	90	Needs improvement
2025-01-16	2025-04-16	INTERN166	Mark Wilson	Male	Medical	Development	90	Good
2025-01-19	2025-04-19	INTERN115	Angela Davis	Male	Medical	Development	90	Needs improvement
2025-01-23	2025-04-23	INTERN048	Brian Thomas	Male	Medical	Development	90	Weak
2025-01-28	2025-04-28	INTERN126	Stephen Watson	Male	Technical Degree	Sales	90	Average
2025-01-30	2025-04-30	INTERN036	Jordan Morales	Female	Medical	Development	90	Nice Work
Total							**23743**	

These visuals helped in clearly identifying patterns and trends in the dataset.

4. Analysis Output and Preparation:

All analysis results, including visuals and insights, were documented and prepared. Screenshots of the visualizations and DAX measures were captured, and the Power BI file was saved. These outputs were uploaded to GitHub as part of Sprint 2 completion and will be used for dashboard development in Sprint 3.